TEMPUS STA LABS

Reference Lab Manual

Setup, hold, skew, multicycle, half-cycle, false-path and clock-gating checks with hand calculations and run results

Package: TEMPUS_STA_LABS_V3_CORRECTED
Library: teaching_sta.lib (constant-delay teaching library, 1 ns time unit)
Tool flow: Cadence Tempus (scripts); results cross-checked with OpenSTA
Version 3.0 - September 2026

Lab	Topic
LAB01	Setup and Hold Checks (4 experiments)
LAB02	Clock Skew (4 experiments)
LAB03	Multicycle Paths (5 experiments)
LAB04	Half-Cycle Paths (3 experiments)
LAB05	False Paths (2 experiments)
LAB06	Clock-Gating Checks (4 experiments)

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1. Introduction

1.1 About this manual

This manual goes with the package **TEMPUS_STA_LABS_V3_CORRECTED**. It contains six self-contained static timing analysis (STA) labs for Cadence Tempus. For every experiment it gives the circuit, the SDC, the formula, a line-by-line hand calculation of BOTH the setup and the hold check, and the result of an actual timing run.

About the run results. The runs quoted in this manual were produced with OpenSTA (open-source STA engine) reading exactly the same Liberty library, netlist and SDC files as the Tempus scripts. Tempus prints the same arrival, required and slack values; only the report layout differs (see section 1.6). Every hand calculation in this manual was checked automatically against these runs; all of them match.

1.2 Package layout

Item	Content
LABxx_NAME/LIB/	teaching_sta.lib - local copy of the teaching library (identical in every lab)
LABxx_NAME/NETLIST/	gate-level Verilog netlist(s) for the lab
LABxx_NAME/SDC/	one independent SDC file per experiment (same base name as the script)
LABxx_NAME/SCRIPTS/	one independent Tempus Tcl script per experiment
LABxx_NAME/REPORTS/	timing reports written by the scripts (.rpt)
run_all_labs.sh	batch runner: one fresh Tempus session per script
README.txt, CHANGES_V3.txt	expected results and list of corrections
LIBRARY_VALUES_AND_CALCULATIONS.txt	plain-text summary of the numbers in this manual

1.3 Running a lab

Always start Tempus from inside the lab folder, and use a new Tempus session for every script:

```
cd LAB02_CLOCK_SKEW
tempus -files SCRIPTS/03_positive_skew_hold_fail.tcl

# when finished, type exit and run the next script

# batch mode (all labs, or selected labs)
./run_all_labs.sh
./run_all_labs.sh LAB03_MULTICYCLE
```

Every script follows the same five steps. Each report is printed on the screen first and then written to ./REPORTS/:

```
if {![file exists ./LIB/teaching_sta.lib]} { puts "ERROR: run from the lab folder";
return }
file mkdir ./REPORTS
read_lib      ./LIB/teaching_sta.lib            ;# 1) library
read_verilog  ./NETLIST/<netlist>.v             ;# 2) netlist
set_top_module <top>                            ;# 3) top module
read_sdc      ./SDC/<experiment>.sdc            ;# 4) constraints
report_timing -late  -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D] -max_paths 1
```

```
report_timing -late  -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D] -max_paths 1
> ./REPORTS/<name>.rpt
                                                ;# 5) -late = setup, -early = hold
```

1.4 Library values (teaching_sta.lib)

All delay and constraint tables in the library are constant, so the value used does not depend on input slew (0.05 ns) or load (0.002 pF). Time unit: 1 ns.

Cell	Liberty arc / check	Symbol	Value (ns)
DFFPOS_X1	CLK -> Q (timing_type rising_edge)	t_clk->q	0.20
DFFPOS_X1	D setup (setup_rising, related_pin CLK)	t_setup	0.20
DFFPOS_X1	D hold (hold_rising, related_pin CLK)	t_hold	0.10
DFFNEG_X1	CLK -> Q (timing_type falling_edge)	t_clk->q	0.20
DFFNEG_X1	D setup / hold (setup_falling / hold_falling)	t_setup / t_hold	0.20 / 0.10
BUF_X1	A -> Y	t_buf	0.50
DLY_X1	A -> Y	t_dly	1.00
INV_X1	A -> Y (not used in the labs)	t_inv	0.40
ICG_X1	EN setup (setup_rising, related_pin CLK)	t_setup,cg	0.50
ICG_X1	EN hold (hold_rising, related_pin CLK)	t_hold,cg	0.20
ICG_X1	CLK -> GCLK	t_icg	0.10

1.5 Timing formulas

General reg-to-reg check

Data arrival time A = launch edge + L_launch + t_clk->q + t_data

Setup required time R_s = capture edge (setup) + L_capture - t_setup - U_setup

Hold required time R_h = capture edge (hold) + L_capture + t_hold + U_hold

Setup slack = R_s - A **Hold slack** = A - R_h (negative = VIOLATED)

Symbol	Meaning
L_launch, L_capture	clock network delay (latency) to the launch / capture flop. 0 for an ideal clock; the sum of clock-cell delays when set_propagated_clock is used
t_data	sum of combinational cell delays between the flops (net delays are 0 in these labs)
U_setup, U_hold	set_clock_uncertainty -setup / -hold
capture edge (setup)	first capture edge after the launch edge; N x T with set_multicycle_path N -setup
capture edge (hold)	setup capture edge - T - H x T, where H is the -hold multicycle value (default 0)
skew	L_capture - L_launch (positive = capture clock later)

Special cases used in the labs

Clock skew (Lab 2): setup slack = (T + L_cap - t_setup) - (L_lau + t_clk->q + t_data); hold slack = (L_lau + t_clk->q + t_data) - (L_cap + t_hold)

Multicycle (Lab 3): R_s = N x T - t_setup; hold edge = N x T - T - H x T (use H = N - 1 to return to 0)

Half cycle (Lab 4): A = T/2 + t_clk->q + t_data; R_s = T - t_setup; T_min = 2 x (t_clk->q + t_data + t_setup)

Clock gating (Lab 6): setup slack = (T - t_setup,cg) - EN_max; hold slack = EN_min - t_hold,cg

Minimum period (single cycle): T_min = t_clk->q + t_data + t_setup + L_lau - L_cap

1.6 Reading a timing report

The same quantities appear in Tempus and OpenSTA reports under different names. Exact Tempus wording can vary slightly between versions.

Tempus report field	OpenSTA report line	Meaning / manual symbol
Path 1: VIOLATED / MET ... Setup|Hold Check	slack (VIOLATED) / (MET)	result of the check
Beginpoint / Endpoint	Startpoint / Endpoint	launch flop / capture pin
Other End Arrival Time	clock ... + clock network lines	capture edge + L_capture
- Setup / + Hold	library setup time / library hold time	t_setup / t_hold
+ Phase Shift	clock CLK (rise edge) value	capture edge (N x T for a multicycle)
- Uncertainty	clock uncertainty	U_setup / U_hold
= Required Time	data required time	R_s / R_h
- Arrival Time	data arrival time	A
= Slack Time	slack	setup or hold slack

1.7 Summary of all results

Lab / experiment	Case	Setup slack	Hold slack	Key idea
LAB01 / 01_setup_fail	FAIL	-0.40	+4.10	T too short
LAB01 / 02_setup_fixed	PASS	+0.60	+4.10	T relaxed
LAB01 / 03_hold_fail	FAIL	+9.60	-0.20	short path + hold uncertainty
LAB01 / 04_hold_fixed	PASS	+9.10	+0.30	buffer in data path
LAB02 / 01_negative_skew_setup_fail	FAIL	-0.40	+5.10	launch clock +1.00
LAB02 / 02_negative_skew_setup_fixed	PASS	+0.10	+4.60	launch clock +0.50
LAB02 / 03_positive_skew_hold_fail	FAIL	+5.60	-0.90	capture clock +1.00
LAB02 / 04_positive_skew_hold_fixed	PASS	+4.60	+0.10	2 buffers in data path
LAB03 / 01_normal_setup_fail	FAIL	-2.40	+6.10	single cycle
LAB03 / 02_mcp2_setup_pass	PASS	+1.60	+2.10	MCP 2, no hold MCP
LAB03 / 03_mcp2_setup_hold_corrected	PASS	+1.60	+6.10	MCP 2 + hold 1
LAB03 / 04_mcp2_short_period_fail	FAIL	-0.40	+6.10	T = 3, MCP 2
LAB03 / 05_mcp3_pass	PASS	+2.60	+6.10	T = 3, MCP 3 + hold 2
LAB04 / 01_half_setup_fail	FAIL	-0.40	+8.10	T = 8, T/2 = 4
LAB04 / 02_half_setup_fixed	PASS	+0.60	+9.10	T = 10, T/2 = 5
LAB04 / 03_half_hold_check	CHECK	+0.60	+9.10	extra T/2 hold margin

Lab / experiment	Case	Setup slack	Hold slack	Key idea
LAB05 / 01_before_false_path	**FAIL**	**cfg: -1.40 data: +1.60**	cfg: +6.10 data: +3.10	cfg path timed
LAB05 / 02_after_false_path	**PASS**	cfg: no path data: +1.60	cfg: no path data: +3.10	cfg path false
LAB06 / 01_cg_setup_pass	**PASS**	+1.50	+7.80	EN at 8.0
LAB06 / 02_cg_setup_fail	**FAIL**	**-0.30**	+9.60	EN at 9.8
LAB06 / 03_cg_setup_fixed	**PASS**	+1.00	+8.30	EN at 8.5
LAB06 / 04_cg_hold_check	**CHECK**	+1.50	+0.30	EN min 0.5

2. Lab 1 - Setup and Hold Checks (LAB01_SETUP_HOLD)

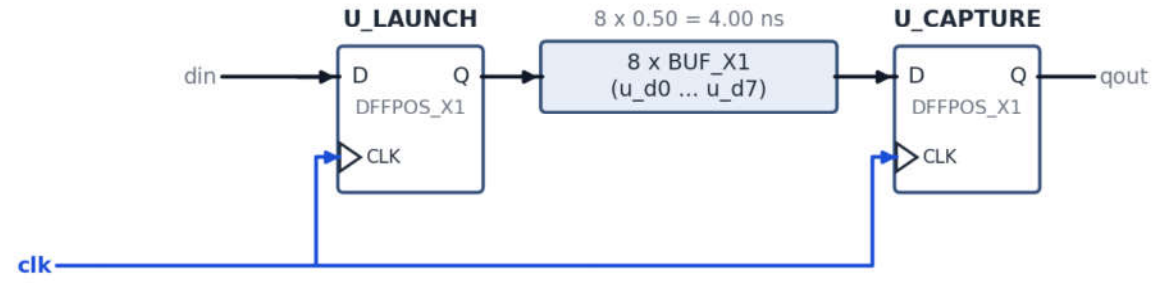
2.1 Objective

- Understand how the setup check limits the clock period and how the hold check protects against a race on the same clock edge.
- Fix a setup violation by relaxing the clock period and a hold violation by adding delay in the data path.

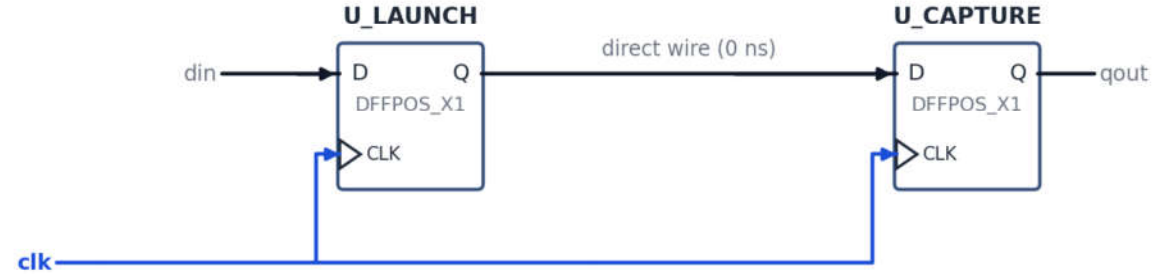
2.2 Files and circuit

File	Description
NETLIST/setup_hold.v	U_LAUNCH -> 8 x BUF_X1 -> U_CAPTURE (4.20 ns path). Used by 01, 02.
NETLIST/hold_fail.v	U_LAUNCH -> direct wire -> U_CAPTURE (0.20 ns path). Used by 03.
NETLIST/hold_fixed.v	U_LAUNCH -> 1 x BUF_X1 -> U_CAPTURE (0.70 ns path). Used by 04.

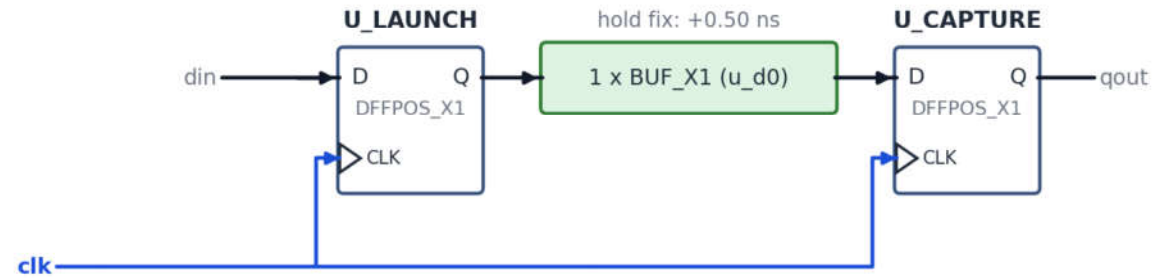
setup_hold.v - used by 01_setup_fail and 02_setup_fixed



hold_fail.v - 03_hold_fail



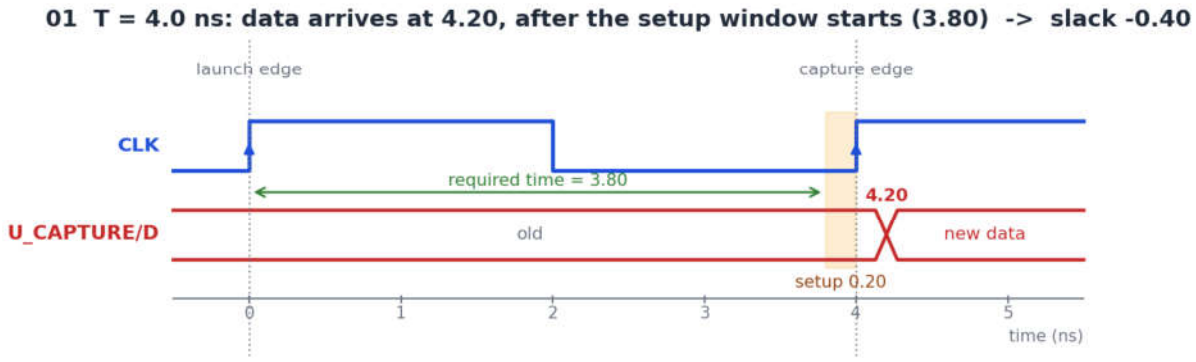
hold_fixed.v - 04_hold_fixed

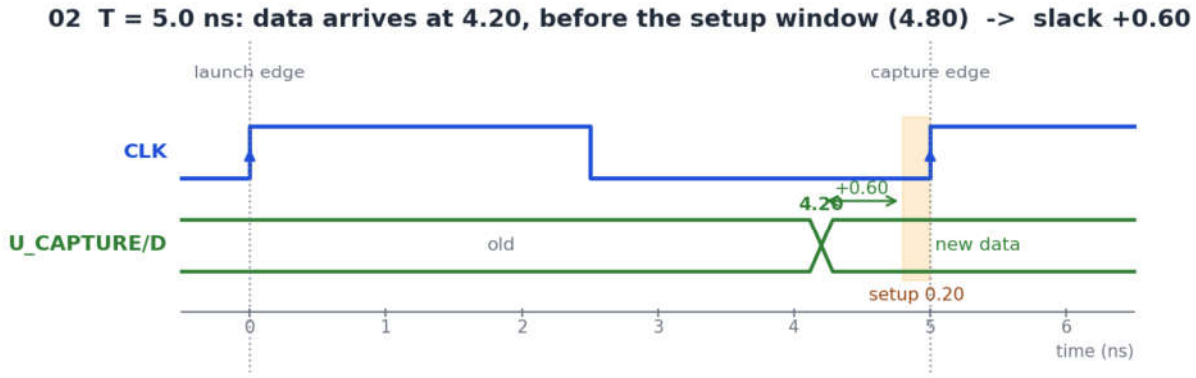


2.3 Concept

Setup asks: does the data launched at one clock edge reach the capture flop at least t_setup before the NEXT clock edge? It depends on the clock period, so a slower clock always helps.

Hold asks: does the NEW data arrive late enough that it does not disturb the value the capture flop is sampling at the SAME clock edge? The clock period does not appear in the hold formula, so only extra data-path delay (or less skew / uncertainty) fixes it.







2.4 Experiment 01_setup_fail [FAIL]

Setup FAIL - clock period too short (T = 4.0 ns)

Item	Value
Run	`cd LAB01_SETUP_HOLD ; tempus -files SCRIPTS/01_setup_fail.tcl`
Netlist / top	`NETLIST/setup_hold.v / setup_hold_top`
SDC	`SDC/01_setup_fail.sdc`
Reports	`REPORTS/01_setup_fail_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 4.0 [get_ports clk]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.20**
Data required time		
Capture clock edge	next rising edge = T	4.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**3.80**
Setup slack	**Required - Arrival = 3.80 - 4.20**	**-0.40 VIOLATED**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.20**
Data required time		
Capture clock edge	same edge (setup edge - T)	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 4.20 - 0.10**	**+4.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	4.20 / 3.80 / -0.40	4.20 / 3.80 / -0.40	VIOLATED	Yes
hold	4.20 / 0.10 / +4.10	4.20 / 0.10 / +4.10	MET	Yes

Why this result

- The path needs CLK->Q + data + setup = 0.20 + 4.00 + 0.20 = 4.40 ns, but the clock gives only 4.00 ns. The data reaches U_CAPTURE/D at 4.20 ns, 0.40 ns after the latest allowed time (3.80 ns).
- Hold is not a problem here: the long 4.20 ns path cannot race through in 0.10 ns.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.00    4.20 ^ U_CAPTURE/D (DFFPOS_X1)
           4.20   data arrival time

   4.00    4.00   clock CLK (rise edge)
   0.00    4.00   clock network delay (ideal)
   0.00    4.00   clock reconvergence pessimism
           4.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    3.80   library setup time
           3.80   data required time
---------------------------------------------------------
           3.80   data required time
          -4.20   data arrival time
---------------------------------------------------------
          -0.40   slack (VIOLATED)
```

2.5 Experiment 02_setup_fixed [PASS]

Setup PASS - clock period relaxed (T = 5.0 ns)

Item	Value
Run	`cd LAB01_SETUP_HOLD ; tempus -files SCRIPTS/02_setup_fixed.tcl`
Netlist / top	`NETLIST/setup_hold.v / setup_hold_top`
SDC	`SDC/02_setup_fixed.sdc`
Reports	`REPORTS/02_setup_fixed_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 4.20**	**+0.60 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.20**
Data required time		
Capture clock edge	same edge (setup edge - T)	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 4.20 - 0.10**	**+4.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	4.20 / 4.80 / +0.60	4.20 / 4.80 / +0.60	MET	Yes
hold	4.20 / 0.10 / +4.10	4.20 / 0.10 / +4.10	MET	Yes

Why this result

- Only the clock period changed. The required time moves from 3.80 to 4.80 ns while the arrival stays 4.20 ns, so the slack becomes +0.60 ns.
- Minimum period for this path: T_min = CLK->Q + data + setup = 4.40 ns (f_max = 1 / 4.40 ns = 227 MHz). The alternative fix is a faster data path: removing one buffer at T = 4.0 gives 3.80 - 3.70 = +0.10 ns.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.00    4.20 ^ U_CAPTURE/D (DFFPOS_X1)
           4.20   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00   clock reconvergence pessimism
           5.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    4.80   library setup time
           4.80   data required time
---------------------------------------------------------
           4.80   data required time
          -4.20   data arrival time
---------------------------------------------------------
           0.60   slack (MET)
```

2.6 Experiment 03_hold_fail [FAIL]

Hold FAIL - short data path plus hold uncertainty

Item	Value
Run	`cd LAB01_SETUP_HOLD ; tempus -files SCRIPTS/03_hold_fail.tcl`
Netlist / top	`NETLIST/hold_fail.v / hold_fail_top`
SDC	`SDC/03_hold_fail.sdc`
Reports	`REPORTS/03_hold_fail_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 10.0 [get_ports clk]
set_clock_uncertainty -hold 0.30 [get_clocks CLK]
```

Hold check (report_timing -early) - main check of this experiment

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold + U_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	direct wire, no cells	0.00
= Data arrival time	**sum of the lines above**	**0.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
+ Hold uncertainty	SDC set_clock_uncertainty -hold	0.30
= Data required time	**sum of the lines above**	**0.40**
Hold slack	**Arrival - Required = 0.20 - 0.40**	**-0.20 VIOLATED**

Setup check (report_timing -late)

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	direct wire, no cells	0.00
= Data arrival time	**sum of the lines above**	**0.20**
Data required time		
Capture clock edge	next rising edge = T	10.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20

Quantity	Where the value comes from	Value (ns)
= Data required time	**sum of the lines above**	**9.80**
Setup slack	**Required - Arrival = 9.80 - 0.20**	**+9.60 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
hold	0.20 / 0.40 / -0.20	0.20 / 0.40 / -0.20	VIOLATED	Yes
setup	0.20 / 9.80 / +9.60	0.20 / 9.80 / +9.60	MET	Yes

Why this result

- Both flops are clocked by the same edge at 0 ns. The capture flop must keep seeing its OLD input until 0.10 (hold) + 0.30 (uncertainty) = 0.40 ns, but the new data from U_LAUNCH arrives at 0.20 ns. The new value races through: slack = 0.20 - 0.40 = -0.20 ns.
- Setup is easy (+9.60 ns): the data is very fast. Changing the period would not help hold because T does not appear in the hold equation.

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.00    0.20 ^ U_CAPTURE/D (DFFPOS_X1)
           0.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.30    0.30   clock uncertainty
   0.00    0.30   clock reconvergence pessimism
           0.30 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.40   library hold time
           0.40   data required time
---------------------------------------------------------
           0.40   data required time
          -0.20   data arrival time
---------------------------------------------------------
          -0.20   slack (VIOLATED)
```

2.7 Experiment 04_hold_fixed [PASS]

Hold PASS - delay buffer inserted in the data path

Item	Value
Run	`cd LAB01_SETUP_HOLD ; tempus -files SCRIPTS/04_hold_fixed.tcl`
Netlist / top	`NETLIST/hold_fixed.v / hold_fixed_top`
SDC	`SDC/04_hold_fixed.sdc`
Reports	`REPORTS/04_hold_fixed_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 10.0 [get_ports clk]
set_clock_uncertainty -hold 0.30 [get_clocks CLK]
```

Hold check (report_timing -early) - main check of this experiment

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold + U_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	1 x BUF_X1 = 1 x 0.50	0.50
= Data arrival time	**sum of the lines above**	**0.70**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
+ Hold uncertainty	SDC set_clock_uncertainty -hold	0.30
= Data required time	**sum of the lines above**	**0.40**
Hold slack	**Arrival - Required = 0.70 - 0.40**	**+0.30 MET**

Setup check (report_timing -late)

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	1 x BUF_X1 = 1 x 0.50	0.50
= Data arrival time	**sum of the lines above**	**0.70**
Data required time		
Capture clock edge	next rising edge = T	10.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20

Quantity	Where the value comes from	Value (ns)
= Data required time	**sum of the lines above**	**9.80**
Setup slack	**Required - Arrival = 9.80 - 0.70**	**+9.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
hold	0.70 / 0.40 / +0.30	0.70 / 0.40 / +0.30	MET	Yes
setup	0.70 / 9.80 / +9.10	0.70 / 9.80 / +9.10	MET	Yes

Why this result

- The SDC is identical to 03. The fix is in the netlist: one BUF_X1 (0.50 ns) delays the new data to 0.70 ns, which is after the 0.40 ns hold window. Slack = +0.30 ns.
- The buffer costs setup margin (9.60 -> 9.10 ns). In real designs, every hold buffer must be re-checked for setup.

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.00    0.70 ^ U_CAPTURE/D (DFFPOS_X1)
           0.70   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.30    0.30   clock uncertainty
   0.00    0.30   clock reconvergence pessimism
           0.30 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.40   library hold time
           0.40   data required time
---------------------------------------------------------
           0.40   data required time
          -0.70   data arrival time
---------------------------------------------------------
           0.30   slack (MET)
```

2.8 Key takeaways

- Setup slack = (T - t_setup) - (t_clk->q + t_data). Increase T or reduce t_data to fix it.
- Hold slack = (t_clk->q + t_data) - (t_hold + uncertainty). T is not in the equation; add data delay to fix it.

2.9 Exercises

1. Find the smallest period for 01 that gives zero setup slack. (Answer: 4.40 ns.)
2. In 03, what hold uncertainty makes the slack exactly 0? (Answer: 0.10 ns.)
3. How many BUF_X1 are needed in 04 if the hold uncertainty is 0.80 ns? (Answer: 2 -> 1.20 - 0.90 = +0.30.)

3. Lab 2 - Clock Skew (LAB02_CLOCK_SKEW)

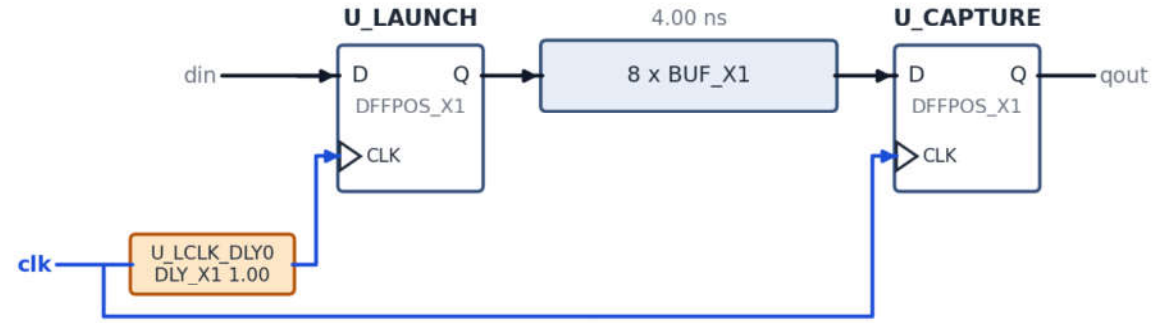
3.1 Objective

- See how clock skew created by real cells in the clock network changes setup and hold.
- Fix negative-skew setup failure by reducing launch clock delay, and positive-skew hold failure by adding data-path delay.

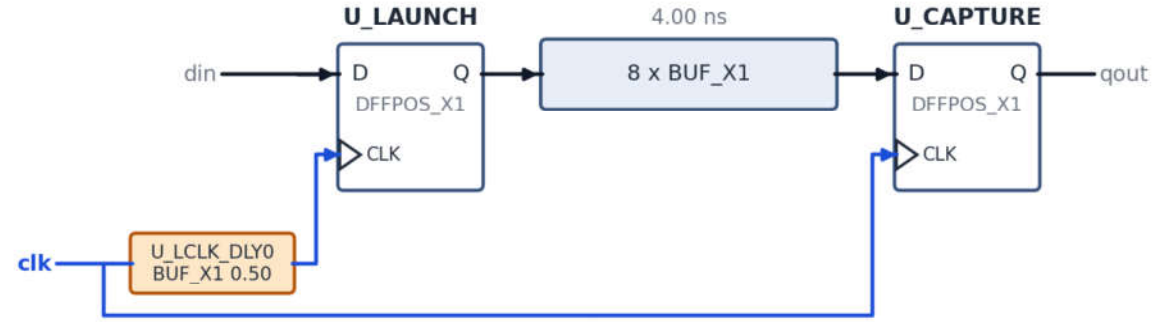
3.2 Files and circuit

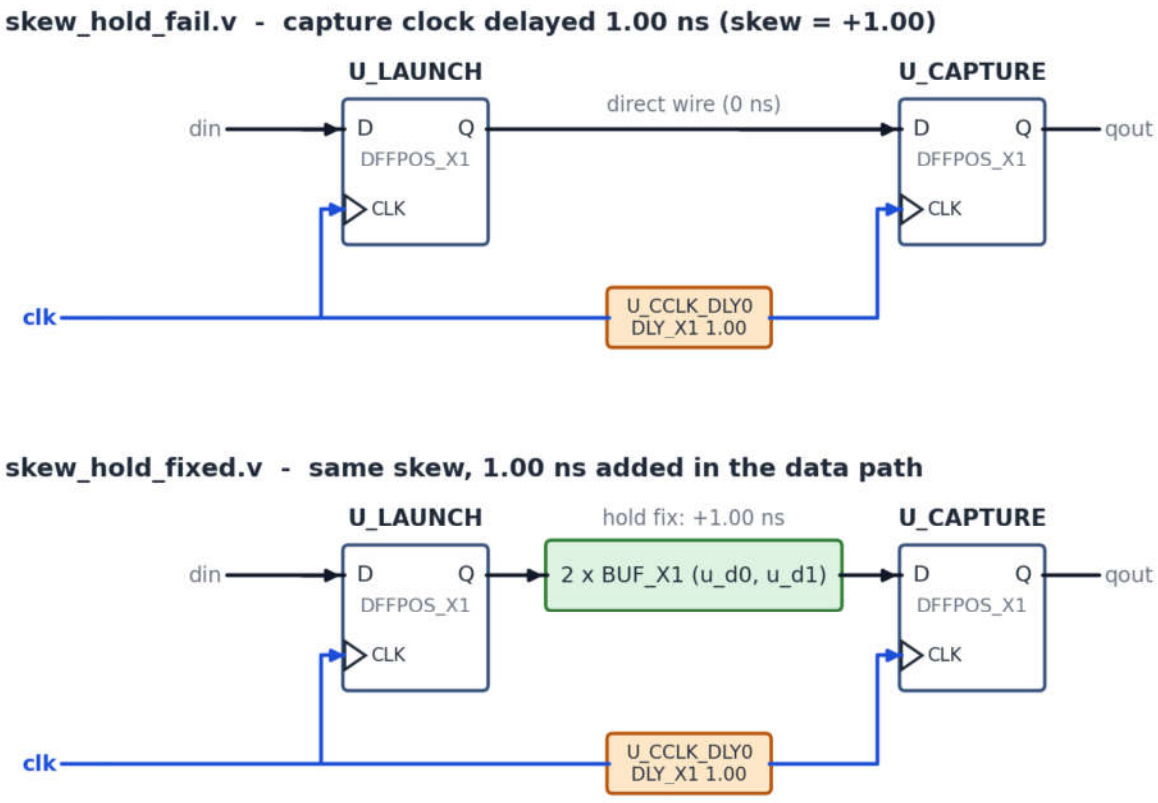
File	Description
NETLIST/skew_setup_fail.v	DLY_X1 (1.00 ns) in the LAUNCH clock branch, 8 x BUF_X1 data path. Used by 01.
NETLIST/skew_setup_fixed.v	BUF_X1 (0.50 ns) in the LAUNCH clock branch, same data path. Used by 02.
NETLIST/skew_hold_fail.v	DLY_X1 (1.00 ns) in the CAPTURE clock branch, direct data wire. Used by 03.
NETLIST/skew_hold_fixed.v	Same capture delay, 2 x BUF_X1 (1.00 ns) in the data path. Used by 04.

skew_setup_fail.v - launch clock delayed 1.00 ns (skew = -1.00)



skew_setup_fixed.v - launch clock delay reduced to 0.50 ns (skew = -0.50)



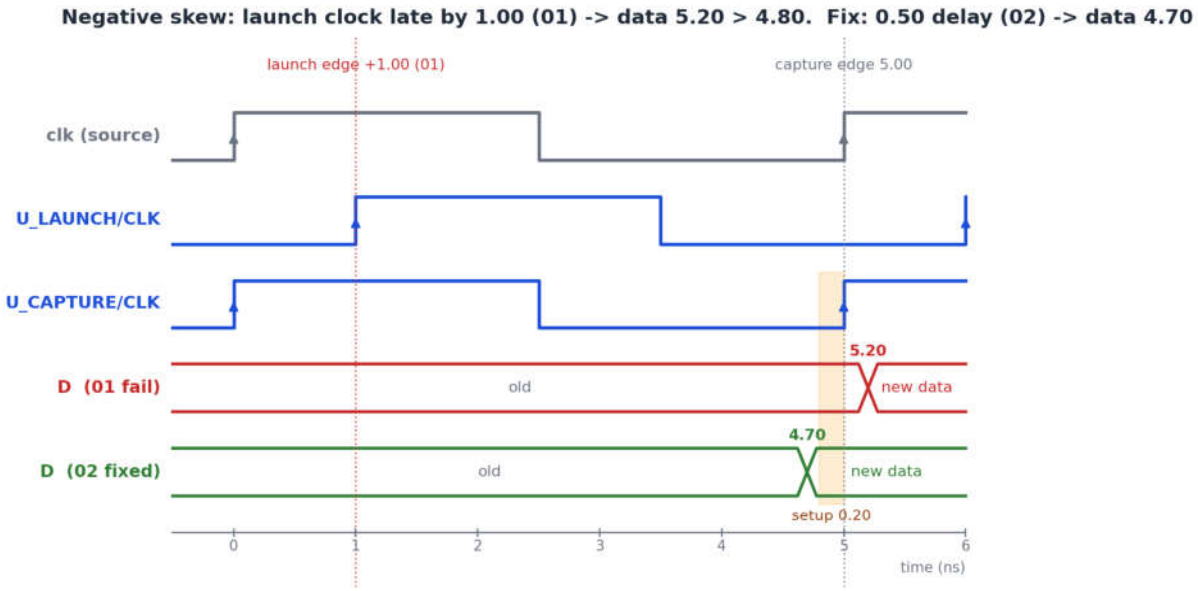


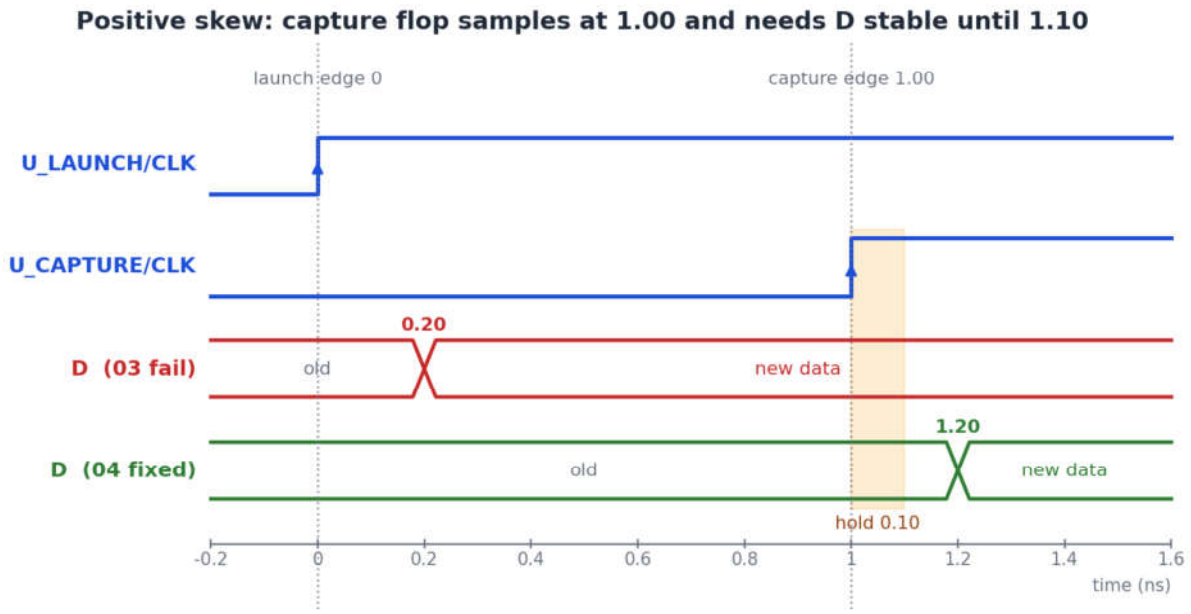
3.3 Concept

Skew = capture clock arrival - launch clock arrival. Negative skew (launch clock later) steals time from setup. Positive skew (capture clock later) makes the hold requirement later, so hold becomes harder.

All four experiments use ONE clock. The skew is created physically by delay cells in the clock branches, so every SDC contains **set_propagated_clock [all_clocks]**. Without it Tempus treats the clock as ideal, ignores U_LCLK_DLY0 / U_CCLK_DLY0 and the skew disappears (this was the V2 bug: hold never failed).

The scripts use **-path_type full_clock** so the clock delay cell is printed line by line in the report.




Positive skew: capture flop samples at 1.00 and needs D stable until 1.10
launch edge 0
capture edge 1.00
U_LAUNCH/CLK
U_CAPTURE/CLK
0.20
D (03 fail)
old
new data
1.20
D (04 fixed)
old
new data
hold 0.10
-0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
time (ns)

3.4 Experiment 01_negative_skew_setup_fail [FAIL]

Setup FAIL - negative skew of -1.00 ns

Item	Value
Run	`cd LAB02_CLOCK_SKEW ; tempus -files SCRIPTS/01_negative_skew_setup_fail.tcl`
Netlist / top	`NETLIST/skew_setup_fail.v / skew_setup_fail_top`
SDC	`SDC/01_negative_skew_setup_fail.sdc`
Reports	`REPORTS/01_negative_skew_setup_fail_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
set_propagated_clock [all_clocks]
```

Clock skew = L_capture - L_launch = 0.00 - 1.00 = -1.00 ns

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	U_LCLK_DLY0 DLY_X1 (propagated)	1.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**5.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	direct clock wire	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 5.20**	**-0.40 VIOLATED**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	U_LCLK_DLY0 DLY_X1 (propagated)	1.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**5.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	direct clock wire	0.00

Quantity	Where the value comes from	Value (ns)
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 5.20 - 0.10**	**+5.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	5.20 / 4.80 / -0.40	5.20 / 4.80 / -0.40	VIOLATED	Yes
hold	5.20 / 0.10 / +5.10	5.20 / 0.10 / +5.10	MET	Yes

Why this result

- The launch flop gets its clock 1.00 ns late, so the data leaves 1.00 ns late, but the capture edge does not move. Arrival 5.20 ns is later than the required 4.80 ns: slack -0.40 ns.
- Without skew the same path would have 5.00 - 0.20 - 4.20 = +0.60 ns. The skew costs exactly 1.00 ns: 0.60 - 1.00 = -0.40.
- Negative skew HELPS hold: +5.10 ns.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   1.00    1.00 ^ U_LCLK_DLY0/Y (DLY_X1)
   0.00    1.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    1.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    1.70 ^ u_d0/Y (BUF_X1)
   0.50    2.20 ^ u_d1/Y (BUF_X1)
   0.50    2.70 ^ u_d2/Y (BUF_X1)
   0.50    3.20 ^ u_d3/Y (BUF_X1)
   0.50    3.70 ^ u_d4/Y (BUF_X1)
   0.50    4.20 ^ u_d5/Y (BUF_X1)
   0.50    4.70 ^ u_d6/Y (BUF_X1)
   0.50    5.20 ^ u_d7/Y (BUF_X1)
   0.00    5.20 ^ U_CAPTURE/D (DFFPOS_X1)
           5.20   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock source latency
   0.00    5.00 ^ clk (in)
   0.00    5.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.00    5.00   clock reconvergence pessimism
  -0.20    4.80   library setup time
           4.80   data required time
---------------------------------------------------------
           4.80   data required time
          -5.20   data arrival time
---------------------------------------------------------
          -0.40   slack (VIOLATED)
```

3.5 Experiment 02_negative_skew_setup_fixed [PASS]

Setup PASS - launch clock delay reduced (skew -0.50 ns)

Item	Value
Run	`cd LAB02_CLOCK_SKEW ; tempus -files SCRIPTS/02_negative_skew_setup_fixed.tcl`
Netlist / top	`NETLIST/skew_setup_fixed.v / skew_setup_fixed_top`
SDC	`SDC/02_negative_skew_setup_fixed.sdc`
Reports	`REPORTS/02_negative_skew_setup_fixed_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
set_propagated_clock [all_clocks]
```

Clock skew = L_capture - L_launch = 0.00 - 0.50 = -0.50 ns

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	U_LCLK_DLY0 BUF_X1 (propagated)	0.50
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.70**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	direct clock wire	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 4.70**	**+0.10 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	U_LCLK_DLY0 BUF_X1 (propagated)	0.50
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**4.70**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	direct clock wire	0.00

Quantity	Where the value comes from	Value (ns)
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 4.70 - 0.10**	**+4.60 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	4.70 / 4.80 / +0.10	4.70 / 4.80 / +0.10	MET	Yes
hold	4.70 / 0.10 / +4.60	4.70 / 0.10 / +4.60	MET	Yes

Why this result

- The clock-tree fix: DLY_X1 (1.00) replaced by BUF_X1 (0.50) in the launch branch. The data leaves 0.50 ns earlier: arrival 4.70 < required 4.80, slack +0.10 ns.
- Other valid fixes: remove one data buffer, add delay in the capture clock branch (useful skew), or increase T.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.50    0.50 ^ U_LCLK_DLY0/Y (BUF_X1)
   0.00    0.50 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.70 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    1.20 ^ u_d0/Y (BUF_X1)
   0.50    1.70 ^ u_d1/Y (BUF_X1)
   0.50    2.20 ^ u_d2/Y (BUF_X1)
   0.50    2.70 ^ u_d3/Y (BUF_X1)
   0.50    3.20 ^ u_d4/Y (BUF_X1)
   0.50    3.70 ^ u_d5/Y (BUF_X1)
   0.50    4.20 ^ u_d6/Y (BUF_X1)
   0.50    4.70 ^ u_d7/Y (BUF_X1)
   0.00    4.70 ^ U_CAPTURE/D (DFFPOS_X1)
           4.70   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock source latency
   0.00    5.00 ^ clk (in)
   0.00    5.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.00    5.00   clock reconvergence pessimism
  -0.20    4.80   library setup time
           4.80   data required time
---------------------------------------------------------
           4.80   data required time
          -4.70   data arrival time
---------------------------------------------------------
           0.10   slack (MET)
```

3.6 Experiment 03_positive_skew_hold_fail [FAIL]

Hold FAIL - positive skew of +1.00 ns

Item	Value
Run	`cd LAB02_CLOCK_SKEW ; tempus -files SCRIPTS/03_positive_skew_hold_fail.tcl`
Netlist / top	`NETLIST/skew_hold_fail.v / skew_hold_fail_top`
SDC	`SDC/03_positive_skew_hold_fail.sdc`
Reports	`REPORTS/03_positive_skew_hold_fail_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
set_propagated_clock [all_clocks]
```

Clock skew = L_capture - L_launch = 1.00 - 0.00 = +1.00 ns

Hold check (report_timing -early) - main check of this experiment

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	direct clock wire	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	direct wire, no cells	0.00
= Data arrival time	**sum of the lines above**	**0.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	U_CCLK_DLY0 DLY_X1 (propagated)	1.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**1.10**
Hold slack	**Arrival - Required = 0.20 - 1.10**	**-0.90 VIOLATED**

Setup check (report_timing -late)

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	direct clock wire	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	direct wire, no cells	0.00
= Data arrival time	**sum of the lines above**	**0.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	U_CCLK_DLY0 DLY_X1 (propagated)	1.00

Quantity	Where the value comes from	Value (ns)
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**5.80**
Setup slack	**Required - Arrival = 5.80 - 0.20**	**+5.60 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
hold	0.20 / 1.10 / -0.90	0.20 / 1.10 / -0.90	VIOLATED	Yes
setup	0.20 / 5.80 / +5.60	0.20 / 5.80 / +5.60	MET	Yes

Why this result

- U_CAPTURE receives the SAME clock edge 1.00 ns late and must see its old input until 1.00 + 0.10 = 1.10 ns. The new data arrives at 0.20 ns and overwrites it: slack -0.90 ns (a race).
- Positive skew HELPS setup: +5.60 ns.

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.00    0.20 ^ U_CAPTURE/D (DFFPOS_X1)
           0.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   1.00    1.00 ^ U_CCLK_DLY0/Y (DLY_X1)
   0.00    1.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.00    1.00   clock reconvergence pessimism
   0.10    1.10   library hold time
           1.10   data required time
---------------------------------------------------------
           1.10   data required time
          -0.20   data arrival time
---------------------------------------------------------
          -0.90   slack (VIOLATED)
```

3.7 Experiment 04_positive_skew_hold_fixed [PASS]

Hold PASS - 2 x BUF_X1 added in the data path

Item	Value
Run	`cd LAB02_CLOCK_SKEW ; tempus -files SCRIPTS/04_positive_skew_hold_fixed.tcl`
Netlist / top	`NETLIST/skew_hold_fixed.v / skew_hold_fixed_top`
SDC	`SDC/04_positive_skew_hold_fixed.sdc`
Reports	`REPORTS/04_positive_skew_hold_fixed_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
set_propagated_clock [all_clocks]
```

Clock skew = L_capture - L_launch = 1.00 - 0.00 = +1.00 ns (unchanged)

Hold check (report_timing -early) - main check of this experiment

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	direct clock wire	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	2 x BUF_X1 = 2 x 0.50	1.00
= Data arrival time	**sum of the lines above**	**1.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	U_CCLK_DLY0 DLY_X1 (propagated)	1.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**1.10**
Hold slack	**Arrival - Required = 1.20 - 1.10**	**+0.10 MET**

Setup check (report_timing -late)

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	direct clock wire	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	2 x BUF_X1 = 2 x 0.50	1.00
= Data arrival time	**sum of the lines above**	**1.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	U_CCLK_DLY0 DLY_X1 (propagated)	1.00

Quantity	Where the value comes from	Value (ns)
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**5.80**
Setup slack	**Required - Arrival = 5.80 - 1.20**	**+4.60 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
hold	1.20 / 1.10 / +0.10	1.20 / 1.10 / +0.10	MET	Yes
setup	1.20 / 5.80 / +4.60	1.20 / 5.80 / +4.60	MET	Yes

Why this result

- Same clock network and same SDC as 03. Two BUF_X1 delay the new data to 1.20 ns, after the 1.10 ns hold requirement: slack +0.10 ns.
- Setup drops from +5.60 to +4.60 ns, still safe. Hold buffers always trade setup margin.

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.00    1.20 ^ U_CAPTURE/D (DFFPOS_X1)
           1.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   1.00    1.00 ^ U_CCLK_DLY0/Y (DLY_X1)
   0.00    1.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.00    1.00   clock reconvergence pessimism
   0.10    1.10   library hold time
           1.10   data required time
---------------------------------------------------------
           1.10   data required time
          -1.20   data arrival time
---------------------------------------------------------
           0.10   slack (MET)
```

3.8 Key takeaways

- Setup slack = (T + L_capture - t_setup) - (L_launch + t_clk->q + t_data)
- Hold slack = (L_launch + t_clk->q + t_data) - (L_capture + t_hold)
- Skew that hurts setup helps hold and vice versa. Physical skew needs set_propagated_clock.

3.9 Exercises

1. In 01, which launch clock delay gives exactly 0 setup slack? (Answer: 0.60 ns.)
2. In 03, instead of buffers, what capture clock delay would make hold pass with +0.10? (Answer: 0.00 ns -> 0.20 - 0.10 = +0.10.)
3. Remove set_propagated_clock from 03 and re-run. Explain the +0.10 ns result.

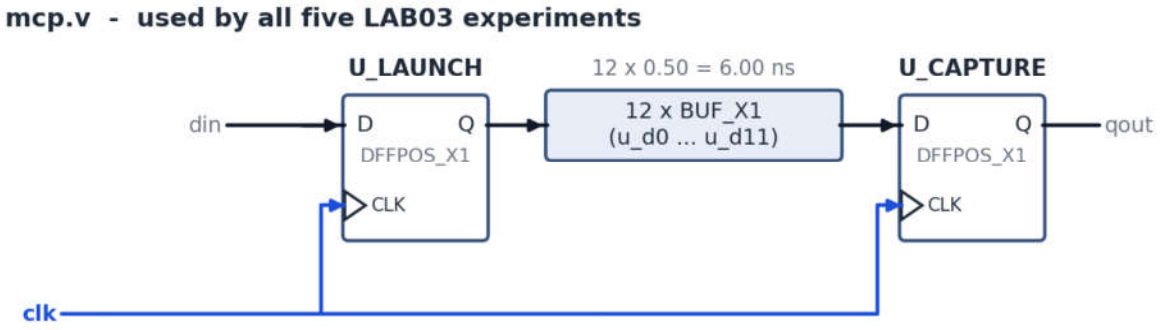
4. Lab 3 - Multicycle Paths (LAB03_MULTICYCLE)

4.1 Objective

- Apply set_multicycle_path to a path that is allowed more than one clock cycle.
- Understand why every setup multicycle N needs a hold multicycle N-1.

4.2 Files and circuit

File	Description
NETLIST/mcp.v	U_LAUNCH -> 12 x BUF_X1 -> U_CAPTURE. Arrival = 0.20 + 6.00 = 6.20 ns. Used by all five experiments.



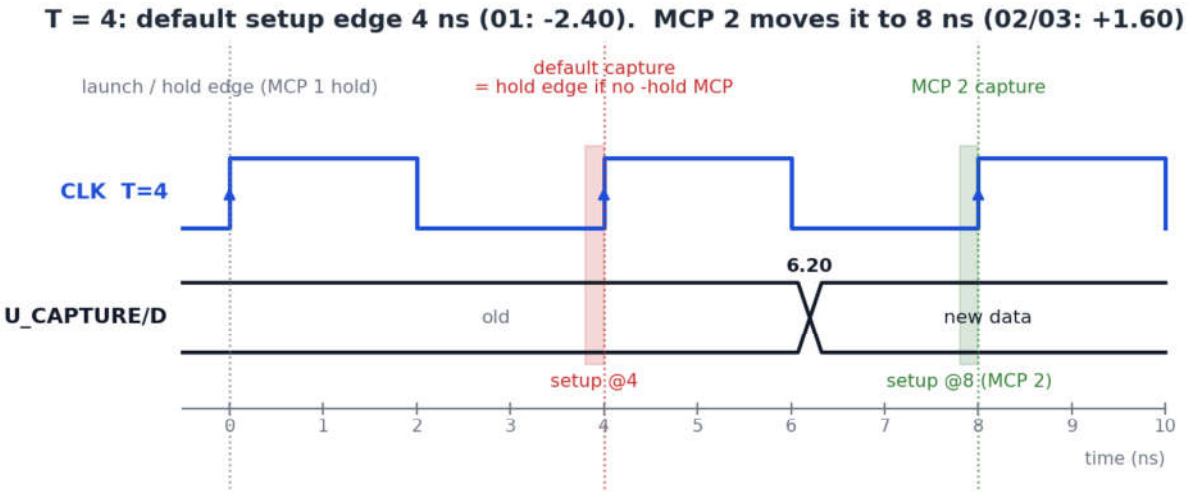
4.3 Concept

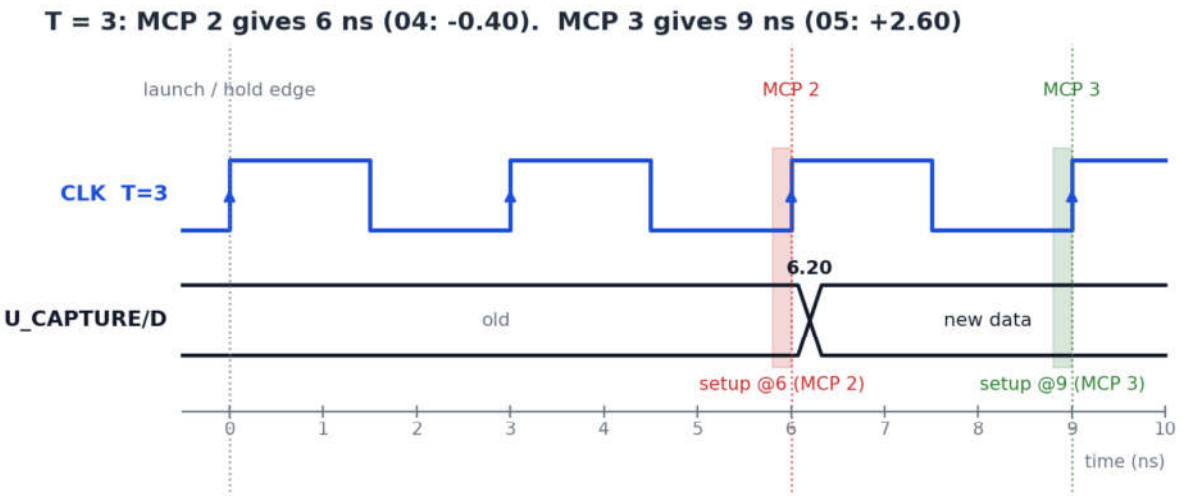
Design assumption: U_CAPTURE only loads new data every N-th clock cycle (for example it has an enable that is active every 2nd cycle). The timing tool cannot see this, so we tell it with set_multicycle_path.

Setup edge with -setup N: capture edge = N x T.

Hold edge = setup edge - T - H x T, where H is the -hold multiplier (default 0). With setup N and no hold exception the hold edge moves to (N-1) x T, which is too pessimistic. Writing -hold N-1 moves it back to 0.

The exception and the report use the same, valid points: -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]. (V2 used U_LAUNCH/Q, which is not a valid start point, so setup always looked like the single-cycle -2.40 ns.)




T = 3: MCP 2 gives 6 ns (04: -0.40). MCP 3 gives 9 ns (05: +2.60)
launch / hold edge
MCP 2
MCP 3
CLK T=3
6.20
U_CAPTURE/D
old
new data
setup @6 (MCP 2)
setup @9 (MCP 3)
0
1
2
3
4
5
6
7
8
9
10
time (ns)

4.4 Experiment 01_normal_setup_fail [FAIL]

Setup FAIL - single-cycle check (T = 4, no exception)

Item	Value
Run	`cd LAB03_MULTICYCLE ; tempus -files SCRIPTS/01_normal_setup_fail.tcl`
Netlist / top	`NETLIST/mcp.v / mcp_top`
SDC	`SDC/01_normal_setup_fail.sdc`
Reports	`REPORTS/01_normal_setup_fail_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 4.0 [get_ports clk]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	1 x T (default)	4.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**3.80**
Setup slack	**Required - Arrival = 3.80 - 6.20**	**-2.40 VIOLATED**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	setup edge - T = 0	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 6.20 - 0.10**	**+6.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	6.20 / 3.80 / -2.40	6.20 / 3.80 / -2.40	VIOLATED	Yes
hold	6.20 / 0.10 / +6.10	6.20 / 0.10 / +6.10	MET	Yes

Why this result

- The 6.20 ns path is timed against a 4.00 ns cycle: 3.80 - 6.20 = -2.40 ns. This is correct if the path really must settle in one cycle.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   4.00    4.00   clock CLK (rise edge)
   0.00    4.00   clock network delay (ideal)
   0.00    4.00   clock reconvergence pessimism
           4.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    3.80   library setup time
           3.80   data required time
---------------------------------------------------------
           3.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
          -2.40   slack (VIOLATED)
```

4.5 Experiment 02_mcp2_setup_pass [PASS]

Setup PASS - MCP 2 setup only (hold edge moves to 4 ns)

Item	Value
Run	`cd LAB03_MULTICYCLE ; tempus -files SCRIPTS/02_mcp2_setup_pass.tcl`
Netlist / top	`NETLIST/mcp.v / mcp_top`
SDC	`SDC/02_mcp2_setup_pass.sdc`
Reports	`REPORTS/02_mcp2_setup_pass_setup.rpt`, `REPORTS/02_mcp2_setup_pass_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 4.0 [get_ports clk]
set_multicycle_path 2 -setup -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	2 x T (MCP 2)	8.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**7.80**
Setup slack	**Required - Arrival = 7.80 - 6.20**	**+1.60 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	8 - 4 - 0 x 4 = 4 (no -hold)	4.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10

Quantity	Where the value comes from	Value (ns)
= Data required time	**sum of the lines above**	**4.10**
Hold slack	**Arrival - Required = 6.20 - 4.10**	**+2.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	6.20 / 7.80 / +1.60	6.20 / 7.80 / +1.60	MET	Yes
hold	6.20 / 4.10 / +2.10	6.20 / 4.10 / +2.10	MET	Yes

Why this result

- Setup now uses the 8 ns edge: 7.80 - 6.20 = +1.60 ns.
- Side effect: the hold edge moved from 0 to 4 ns, so hold required = 4.10 ns. It still passes (+2.10) only because this path is long; a path shorter than 3.90 ns would show a false hold violation. Compare with 03.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   8.00    8.00   clock CLK (rise edge)
   0.00    8.00   clock network delay (ideal)
   0.00    8.00   clock reconvergence pessimism
           8.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    7.80   library setup time
           7.80   data required time
---------------------------------------------------------
           7.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
           1.60   slack (MET)
```

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
```

```
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   4.00    4.00   clock CLK (rise edge)
   0.00    4.00   clock network delay (ideal)
   0.00    4.00   clock reconvergence pessimism
           4.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    4.10   library hold time
           4.10   data required time
---------------------------------------------------------
           4.10   data required time
          -6.20   data arrival time
---------------------------------------------------------
           2.10   slack (MET)
```

4.6 Experiment 03_mcp2_setup_hold_corrected [PASS]

PASS - MCP 2 setup with companion MCP 1 hold

Item	Value
Run	`cd LAB03_MULTICYCLE ; tempus -files SCRIPTS/03_mcp2_setup_hold_corrected.tcl`
Netlist / top	`NETLIST/mcp.v / mcp_top`
SDC	`SDC/03_mcp2_setup_hold_corrected.sdc`
Reports	`REPORTS/03_mcp2_setup_hold_corrected_setup.rpt`, `REPORTS/03_mcp2_setup_hold_corrected_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 4.0 [get_ports clk]
set_multicycle_path 2 -setup -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
set_multicycle_path 1 -hold  -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	2 x T (MCP 2)	8.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**7.80**
Setup slack	**Required - Arrival = 7.80 - 6.20**	**+1.60 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	8 - 4 - 1 x 4 = 0 (MCP 1 hold)	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00

Quantity	Where the value comes from	Value (ns)
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 6.20 - 0.10**	**+6.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	6.20 / 7.80 / +1.60	6.20 / 7.80 / +1.60	MET	Yes
hold	6.20 / 0.10 / +6.10	6.20 / 0.10 / +6.10	MET	Yes

Why this result

- The correct pair. Setup is unchanged (+1.60 ns) and the hold edge returns to 0 ns: 6.20 - 0.10 = +6.10 ns.
- Rule: set_multicycle_path N -setup -> set_multicycle_path N-1 -hold.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   8.00    8.00   clock CLK (rise edge)
   0.00    8.00   clock network delay (ideal)
   0.00    8.00   clock reconvergence pessimism
           8.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    7.80   library setup time
           7.80   data required time
---------------------------------------------------------
           7.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
           1.60   slack (MET)
```

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
```

```
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.10   library hold time
           0.10   data required time
---------------------------------------------------------
           0.10   data required time
          -6.20   data arrival time
---------------------------------------------------------
           6.10   slack (MET)
```

4.7 Experiment 04_mcp2_short_period_fail [FAIL]

Setup FAIL - MCP 2 is not enough at T = 3

Item	Value
Run	`cd LAB03_MULTICYCLE ; tempus -files SCRIPTS/04_mcp2_short_period_fail.tcl`
Netlist / top	`NETLIST/mcp.v / mcp_top`
SDC	`SDC/04_mcp2_short_period_fail.sdc`
Reports	`REPORTS/04_mcp2_short_period_fail_setup.rpt, REPORTS/04_mcp2_short_period_fail_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 3.0 [get_ports clk]
set_multicycle_path 2 -setup -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
set_multicycle_path 1 -hold  -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	2 x T (MCP 2)	6.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**5.80**
Setup slack	**Required - Arrival = 5.80 - 6.20**	**-0.40 VIOLATED**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	6 - 3 - 1 x 3 = 0	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00

Quantity	Where the value comes from	Value (ns)
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 6.20 - 0.10**	**+6.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	6.20 / 5.80 / -0.40	6.20 / 5.80 / -0.40	VIOLATED	Yes
hold	6.20 / 0.10 / +6.10	6.20 / 0.10 / +6.10	MET	Yes

Why this result

- A multicycle exception only gives N x T. Two 3 ns cycles = 6 ns, but the path needs 6.20 + 0.20 = 6.40 ns: slack -0.40 ns.
- Minimum N for T = 3: N x 3 >= 6.40 -> N = 3.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   6.00    6.00   clock CLK (rise edge)
   0.00    6.00   clock network delay (ideal)
   0.00    6.00   clock reconvergence pessimism
           6.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    5.80   library setup time
           5.80   data required time
---------------------------------------------------------
           5.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
          -0.40   slack (VIOLATED)
```

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
```

```
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.10   library hold time
           0.10   data required time
---------------------------------------------------------
           0.10   data required time
          -6.20   data arrival time
---------------------------------------------------------
           6.10   slack (MET)
```

4.8 Experiment 05_mcp3_pass [PASS]

PASS - MCP 3 setup with MCP 2 hold at T = 3

Item	Value
Run	`cd LAB03_MULTICYCLE ; tempus -files SCRIPTS/05_mcp3_pass.tcl`
Netlist / top	`NETLIST/mcp.v / mcp_top`
SDC	`SDC/05_mcp3_pass.sdc`
Reports	`REPORTS/05_mcp3_pass_setup.rpt, REPORTS/05_mcp3_pass_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 3.0 [get_ports clk]
set_multicycle_path 3 -setup -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
set_multicycle_path 2 -hold  -from [get_pins U_LAUNCH/CLK] -to [get_pins U_CAPTURE/D]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	3 x T (MCP 3)	9.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**8.80**
Setup slack	**Required - Arrival = 8.80 - 6.20**	**+2.60 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	9 - 3 - 2 x 3 = 0 (MCP 2 hold)	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10

Quantity	Where the value comes from	Value (ns)
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 6.20 - 0.10**	**+6.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	6.20 / 8.80 / +2.60	6.20 / 8.80 / +2.60	MET	Yes
hold	6.20 / 0.10 / +6.10	6.20 / 0.10 / +6.10	MET	Yes

Why this result

- Three cycles give 9 ns: 8.80 - 6.20 = +2.60 ns. The -hold 2 companion keeps the hold edge at 0: +6.10 ns.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   9.00    9.00   clock CLK (rise edge)
   0.00    9.00   clock network delay (ideal)
   0.00    9.00   clock reconvergence pessimism
           9.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    8.80   library setup time
           8.80   data required time
---------------------------------------------------------
           8.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
           2.60   slack (MET)
```

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min
```

```
  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_LAUNCH/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_LAUNCH/Q (DFFPOS_X1)
   0.50    0.70 ^ u_d0/Y (BUF_X1)
   0.50    1.20 ^ u_d1/Y (BUF_X1)
   0.50    1.70 ^ u_d2/Y (BUF_X1)
   0.50    2.20 ^ u_d3/Y (BUF_X1)
   0.50    2.70 ^ u_d4/Y (BUF_X1)
   0.50    3.20 ^ u_d5/Y (BUF_X1)
   0.50    3.70 ^ u_d6/Y (BUF_X1)
   0.50    4.20 ^ u_d7/Y (BUF_X1)
   0.50    4.70 ^ u_d8/Y (BUF_X1)
   0.50    5.20 ^ u_d9/Y (BUF_X1)
   0.50    5.70 ^ u_d10/Y (BUF_X1)
   0.50    6.20 ^ u_d11/Y (BUF_X1)
   0.00    6.20 ^ U_CAPTURE/D (DFFPOS_X1)
           6.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.10   library hold time
           0.10   data required time
---------------------------------------------------------
           0.10   data required time
          -6.20   data arrival time
---------------------------------------------------------
           6.10   slack (MET)
```

4.9 Key takeaways

- Setup required = N x T - t_setup. Hold edge = N x T - T - H x T.
- Always write the pair: -setup N and -hold N-1.
- Check the capture edge in the report ('Phase Shift' in Tempus) to prove the exception was applied.

4.10 Exercises

1. In 02, shorten the data path to 6 buffers (3.20 ns). What is the hold slack without and with -hold 1? (Answer: 3.20 - 4.10 = -0.90; 3.20 - 0.10 = +3.10.)
2. What is the smallest T that passes with MCP 2? (Answer: 3.20 ns.)

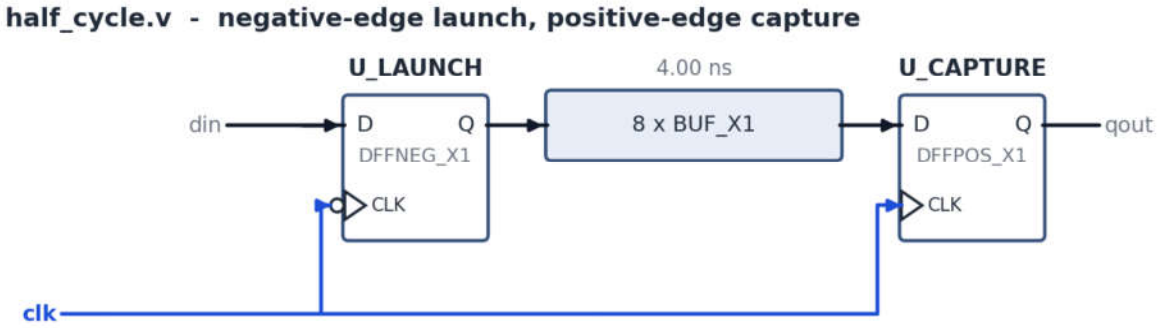
5. Lab 4 - Half-Cycle Paths (LAB04_HALF_CYCLE)

5.1 Objective

- Time a path launched on the falling edge and captured on the rising edge.
- See why such a path gets only half the period for setup and extra margin for hold.

5.2 Files and circuit

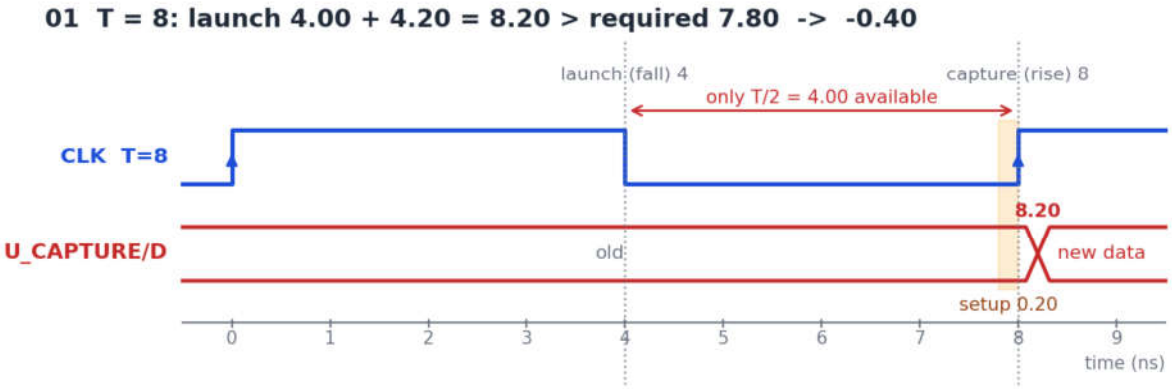
File	Description
NETLIST/half_cycle.v	U_LAUNCH (DFFNEG_X1) -> 8 x BUF_X1 -> U_CAPTURE (DFFPOS_X1). Data delay 4.20 ns.

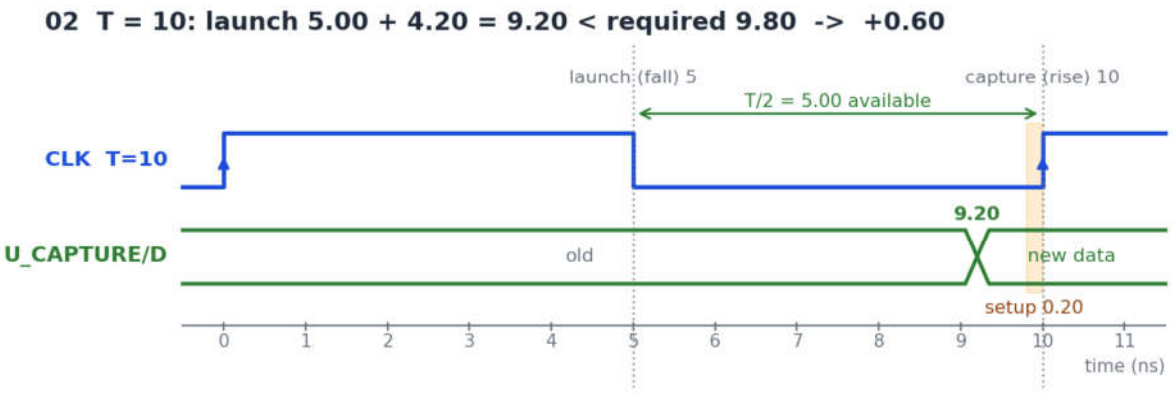


5.3 Concept

The launch flop fires on the FALLING edge (T/2); the capture flop samples on the next RISING edge (T). Setup therefore has only T/2 available.

For hold, the capture edge is the rising edge one period before the setup capture edge (T - T = 0). The data was launched at T/2, so hold gets an extra half period of margin.




02 T = 10: launch 5.00 + 4.20 = 9.20 < required 9.80 -> +0.60
launch (fall) 5
capture (rise) 10
T/2 = 5.00 available
CLK T=10
9.20
U_CAPTURE/D
old
new data
setup 0.20
0
1
2
3
4
5
6
7
8
9
10
11
time (ns)

5.4 Experiment 01_half_setup_fail [FAIL]

Setup FAIL - T = 8 ns, only 4 ns available

Item	Value
Run	`cd LAB04_HALF_CYCLE ; tempus -files SCRIPTS/01_half_setup_fail.tcl`
Netlist / top	`NETLIST/half_cycle.v / half_cycle_top`
SDC	`SDC/01_half_setup_fail.sdc`
Reports	`REPORTS/01_half_setup_fail_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 8.0 -waveform {0 4} [get_ports clk]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	4.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**8.20**
Data required time		
Capture clock edge	next rising edge = T	8.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**7.80**
Setup slack	**Required - Arrival = 7.80 - 8.20**	**-0.40 VIOLATED**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	4.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**8.20**
Data required time		
Capture clock edge	setup edge - T = 0	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 8.20 - 0.10**	**+8.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	8.20 / 7.80 / -0.40	8.20 / 7.80 / -0.40	VIOLATED	Yes
hold	8.20 / 0.10 / +8.10	8.20 / 0.10 / +8.10	MET	Yes

Why this result

- Arrival = 4.00 + 0.20 + 4.00 = 8.20 ns, required = 8.00 - 0.20 = 7.80 ns: slack -0.40 ns.
- The same path between two rising-edge flops would pass at T = 8 with +3.60 ns. The half cycle costs T/2.

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (falling edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   4.00    4.00   clock CLK (fall edge)
   0.00    4.00   clock network delay (ideal)
   0.00    4.00 v U_LAUNCH/CLK (DFFNEG_X1)
   0.20    4.20 ^ U_LAUNCH/Q (DFFNEG_X1)
   0.50    4.70 ^ u_d0/Y (BUF_X1)
   0.50    5.20 ^ u_d1/Y (BUF_X1)
   0.50    5.70 ^ u_d2/Y (BUF_X1)
   0.50    6.20 ^ u_d3/Y (BUF_X1)
   0.50    6.70 ^ u_d4/Y (BUF_X1)
   0.50    7.20 ^ u_d5/Y (BUF_X1)
   0.50    7.70 ^ u_d6/Y (BUF_X1)
   0.50    8.20 ^ u_d7/Y (BUF_X1)
   0.00    8.20 ^ U_CAPTURE/D (DFFPOS_X1)
           8.20   data arrival time

   8.00    8.00   clock CLK (rise edge)
   0.00    8.00   clock network delay (ideal)
   0.00    8.00   clock reconvergence pessimism
           8.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    7.80   library setup time
           7.80   data required time
---------------------------------------------------------
           7.80   data required time
          -8.20   data arrival time
---------------------------------------------------------
          -0.40   slack (VIOLATED)
```

5.5 Experiment 02_half_setup_fixed [PASS]

Setup PASS - T = 10 ns, 5 ns available

Item	Value
Run	`cd LAB04_HALF_CYCLE ; tempus -files SCRIPTS/02_half_setup_fixed.tcl`
Netlist / top	`NETLIST/half_cycle.v / half_cycle_top`
SDC	`SDC/02_half_setup_fixed.sdc`
Reports	`REPORTS/02_half_setup_fixed_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 10.0 -waveform {0 5} [get_ports clk]
```

Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	5.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**9.20**
Data required time		
Capture clock edge	next rising edge = T	10.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**9.80**
Setup slack	**Required - Arrival = 9.80 - 9.20**	**+0.60 MET**

Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	5.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**9.20**
Data required time		
Capture clock edge	setup edge - T = 0	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 9.20 - 0.10**	**+9.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
setup	9.20 / 9.80 / +0.60	9.20 / 9.80 / +0.60	MET	Yes
hold	9.20 / 0.10 / +9.10	9.20 / 0.10 / +9.10	MET	Yes

Why this result

- Arrival = 5.00 + 4.20 = 9.20 ns, required = 9.80 ns: slack +0.60 ns.
- Minimum period: T/2 >= 4.40 -> T_min = 8.80 ns (twice the 4.40 ns a full-cycle path needs).

Run report - setup (max path, reference run)

```
Startpoint: U_LAUNCH (falling edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   5.00    5.00   clock CLK (fall edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00 v U_LAUNCH/CLK (DFFNEG_X1)
   0.20    5.20 ^ U_LAUNCH/Q (DFFNEG_X1)
   0.50    5.70 ^ u_d0/Y (BUF_X1)
   0.50    6.20 ^ u_d1/Y (BUF_X1)
   0.50    6.70 ^ u_d2/Y (BUF_X1)
   0.50    7.20 ^ u_d3/Y (BUF_X1)
   0.50    7.70 ^ u_d4/Y (BUF_X1)
   0.50    8.20 ^ u_d5/Y (BUF_X1)
   0.50    8.70 ^ u_d6/Y (BUF_X1)
   0.50    9.20 ^ u_d7/Y (BUF_X1)
   0.00    9.20 ^ U_CAPTURE/D (DFFPOS_X1)
           9.20   data arrival time

  10.00   10.00   clock CLK (rise edge)
   0.00   10.00   clock network delay (ideal)
   0.00   10.00   clock reconvergence pessimism
          10.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
  -0.20    9.80   library setup time
           9.80   data required time
---------------------------------------------------------
           9.80   data required time
          -9.20   data arrival time
---------------------------------------------------------
           0.60   slack (MET)
```

5.6 Experiment 03_half_hold_check [CHECK]

Hold check - large margin on a half-cycle path

Item	Value
Run	`cd LAB04_HALF_CYCLE ; tempus -files SCRIPTS/03_half_hold_check.tcl`
Netlist / top	`NETLIST/half_cycle.v / half_cycle_top`
SDC	`SDC/03_half_hold_check.sdc`
Reports	`REPORTS/03_half_hold_check_hold.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 10.0 -waveform {0 5} [get_ports clk]
```

Hold check (report_timing -early) - main check of this experiment

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	5.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**9.20**
Data required time		
Capture clock edge	setup edge - T = 0	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 9.20 - 0.10**	**+9.10 MET**

Setup check (report_timing -late)

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	falling edge = T/2	5.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFNEG_X1: CLK->Q (falling_edge)	0.20
+ Data path delay	8 x BUF_X1 = 8 x 0.50	4.00
= Data arrival time	**sum of the lines above**	**9.20**
Data required time		
Capture clock edge	next rising edge = T	10.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**9.80**
Setup slack	**Required - Arrival = 9.80 - 9.20**	**+0.60 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
hold	9.20 / 0.10 / +9.10	9.20 / 0.10 / +9.10	**MET**	**Yes**
setup	9.20 / 9.80 / +0.60	9.20 / 9.80 / +0.60	**MET**	**Yes**

Why this result

- Hold compares the data launched at 5 ns with the rising edge at 0 ns: 9.20 - 0.10 = +9.10 ns. Equivalently, the next launch (15 ns) against the setup edge (10 ns) gives (15 + 4.20) - (10 + 0.10) = +9.10 ns.
- Because of this extra T/2, a falling-to-rising path almost never fails hold, so this lab has no hold-fail case.

Run report - hold (min path, reference run)

```
Startpoint: U_LAUNCH (falling edge-triggered flip-flop clocked by CLK)
Endpoint: U_CAPTURE (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   5.00    5.00   clock CLK (fall edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00 v U_LAUNCH/CLK (DFFNEG_X1)
   0.20    5.20 ^ U_LAUNCH/Q (DFFNEG_X1)
   0.50    5.70 ^ u_d0/Y (BUF_X1)
   0.50    6.20 ^ u_d1/Y (BUF_X1)
   0.50    6.70 ^ u_d2/Y (BUF_X1)
   0.50    7.20 ^ u_d3/Y (BUF_X1)
   0.50    7.70 ^ u_d4/Y (BUF_X1)
   0.50    8.20 ^ u_d5/Y (BUF_X1)
   0.50    8.70 ^ u_d6/Y (BUF_X1)
   0.50    9.20 ^ u_d7/Y (BUF_X1)
   0.00    9.20 ^ U_CAPTURE/D (DFFPOS_X1)
           9.20   data arrival time

   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ U_CAPTURE/CLK (DFFPOS_X1)
   0.10    0.10   library hold time
           0.10   data required time
---------------------------------------------------------
           0.10   data required time
          -9.20   data arrival time
---------------------------------------------------------
           9.10   slack (MET)
```

5.7 Key takeaways

- Setup slack = T - t_setup - (T/2 + t_clk->q + t_data) -> T_min = 2 x (t_clk->q + t_data + t_setup).
- Hold slack = (T/2 + t_clk->q + t_data) - t_hold: large by construction.

5.8 Exercises

1. With T = 10, how many buffers can the path have before setup fails? (Answer: 9 -> 5 + 4.70 = 9.70 < 9.80; 10 buffers -> 10.20 > 9.80.)
2. Change U_LAUNCH to DFFPOS_X1 at T = 8. What is the setup slack? (Answer: 7.80 - 4.20 = +3.60.)

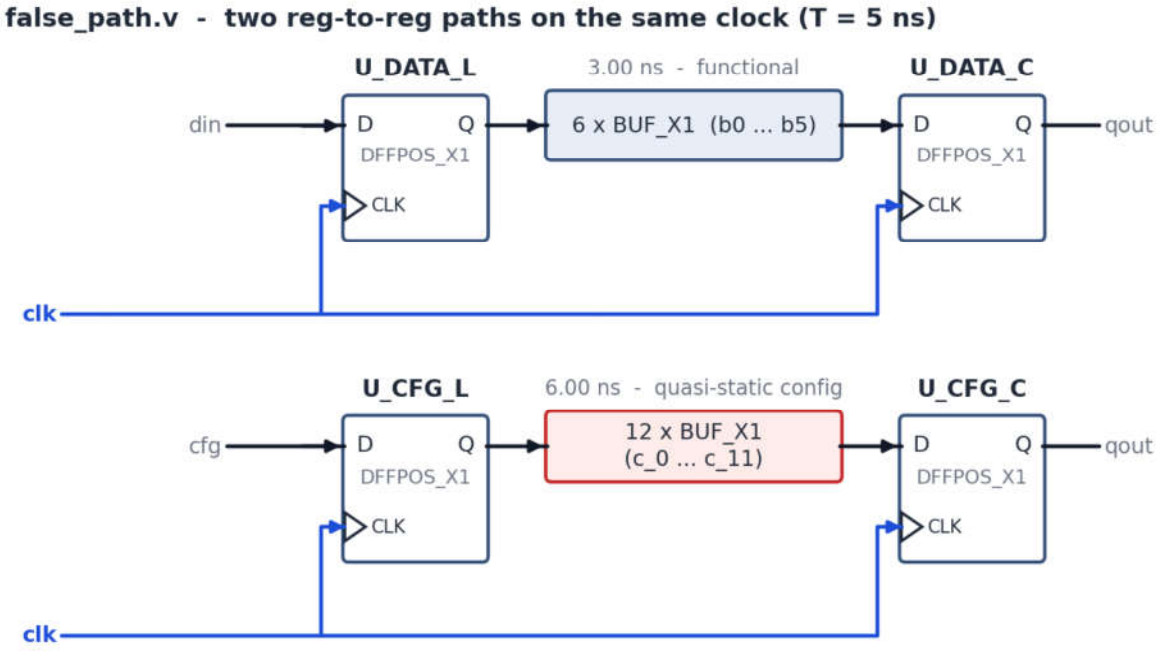
6. Lab 5 - False Paths (LAB05_FALSE_PATH)

6.1 Objective

- Remove a path that does not need timing (a quasi-static configuration path) with set_false_path.
- Verify that the exception removes only the intended path.

6.2 Files and circuit

File	Description
NETLIST/false_path.v	Data path U_DATA_L -> 6 x BUF_X1 -> U_DATA_C (3.20 ns); config path U_CFG_L -> 12 x BUF_X1 -> U_CFG_C (6.20 ns).



6.3 Concept

Assumption: the cfg register is written once (for example after reset) and is stable during normal operation. Its path never has to settle within one clock cycle, but the timing tool times every reg-to-reg path unless told otherwise.

set_false_path removes the path from BOTH setup and hold analysis. It does not change the circuit; it changes what the tool checks. A wrong false path hides a real bug, so it must be justified by the design.

The exception must use valid start/end points: -from [get_pins U_CFG_L/CLK] -to [get_pins U_CFG_C/D]. In V2, -from U_CFG_L/Q was ignored with a warning and the path stayed at -1.40 ns.

6.4 Experiment 01_before_false_path [FAIL]

FAIL - config path timed as a normal single-cycle path

Item	Value
Run	`cd LAB05_FALSE_PATH ; tempus -files SCRIPTS/01_before_false_path.tcl`
Netlist / top	`NETLIST/false_path.v / false_path_top`
SDC	`SDC/01_before_false_path.sdc`
Reports	`REPORTS/01_before_false_path_cfg_setup.rpt`, `REPORTS/01_before_false_path_data_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
```

Config path U_CFG_L -> U_CFG_C - Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 6.20**	**-1.40 VIOLATED**

Config path U_CFG_L -> U_CFG_C - Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	12 x BUF_X1 = 12 x 0.50	6.00
= Data arrival time	**sum of the lines above**	**6.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10

Quantity	Where the value comes from	Value (ns)
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 6.20 - 0.10**	**+6.10 MET**

Data path U_DATA_L -> U_DATA_C - Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	6 x BUF_X1 = 6 x 0.50	3.00
= Data arrival time	**sum of the lines above**	**3.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 3.20**	**+1.60 MET**

Data path U_DATA_L -> U_DATA_C - Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	6 x BUF_X1 = 6 x 0.50	3.00
= Data arrival time	**sum of the lines above**	**3.20**
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 3.20 - 0.10**	**+3.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
cfg - setup	6.20 / 4.80 / -1.40	6.20 / 4.80 / -1.40	VIOLATED	Yes
cfg - hold	6.20 / 0.10 / +6.10	6.20 / 0.10 / +6.10	MET	Yes
data - setup	3.20 / 4.80 / +1.60	3.20 / 4.80 / +1.60	MET	Yes

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
data - hold	3.20 / 0.10 / +3.10	3.20 / 0.10 / +3.10	MET	Yes

Why this result

- Config path: 4.80 - 6.20 = -1.40 ns. The functional data path passes with +1.60 ns.
- The -1.40 ns violation is not a real functional problem under the lab assumption, but the tool cannot know that.

Run report - cfg path, setup (max path, reference run)

```
Startpoint: U_CFG_L (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_CFG_C (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_CFG_L/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_CFG_L/Q (DFFPOS_X1)
   0.50    0.70 ^ c_0/Y (BUF_X1)
   0.50    1.20 ^ c_1/Y (BUF_X1)
   0.50    1.70 ^ c_2/Y (BUF_X1)
   0.50    2.20 ^ c_3/Y (BUF_X1)
   0.50    2.70 ^ c_4/Y (BUF_X1)
   0.50    3.20 ^ c_5/Y (BUF_X1)
   0.50    3.70 ^ c_6/Y (BUF_X1)
   0.50    4.20 ^ c_7/Y (BUF_X1)
   0.50    4.70 ^ c_8/Y (BUF_X1)
   0.50    5.20 ^ c_9/Y (BUF_X1)
   0.50    5.70 ^ c_10/Y (BUF_X1)
   0.50    6.20 ^ c_11/Y (BUF_X1)
   0.00    6.20 ^ U_CFG_C/D (DFFPOS_X1)
           6.20   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00   clock reconvergence pessimism
           5.00 ^ U_CFG_C/CLK (DFFPOS_X1)
  -0.20    4.80   library setup time
           4.80   data required time
---------------------------------------------------------
           4.80   data required time
          -6.20   data arrival time
---------------------------------------------------------
          -1.40   slack (VIOLATED)
```

Run report - data path, setup (max path, reference run)

```
Startpoint: U_DATA_L (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_DATA_C (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_DATA_L/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_DATA_L/Q (DFFPOS_X1)
   0.50    0.70 ^ b0/Y (BUF_X1)
```

```
   0.50    1.20 ^ b1/Y (BUF_X1)
   0.50    1.70 ^ b2/Y (BUF_X1)
   0.50    2.20 ^ b3/Y (BUF_X1)
   0.50    2.70 ^ b4/Y (BUF_X1)
   0.50    3.20 ^ b5/Y (BUF_X1)
   0.00    3.20 ^ U_DATA_C/D (DFFPOS_X1)
           3.20   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00   clock reconvergence pessimism
           5.00 ^ U_DATA_C/CLK (DFFPOS_X1)
  -0.20    4.80   library setup time
           4.80   data required time
-----------------------------------------------------------
           4.80   data required time
          -3.20   data arrival time
-----------------------------------------------------------
           1.60   slack (MET)
```

6.5 Experiment 02_after_false_path [PASS]

PASS - config path declared false

Item	Value
Run	`cd LAB05_FALSE_PATH ; tempus -files SCRIPTS/02_after_false_path.tcl`
Netlist / top	`NETLIST/false_path.v / false_path_top`
SDC	`SDC/02_after_false_path.sdc`
Reports	`REPORTS/02_after_false_path_cfg_setup.rpt`, `REPORTS/02_after_false_path_top5_setup.rpt`

Constraints (SDC)

```
create_clock -name CLK -period 5.0 [get_ports clk]
set_false_path -from [get_pins U_CFG_L/CLK] -to [get_pins U_CFG_C/D]
```

Config path U_CFG_L -> U_CFG_C: not timed

> set_false_path removes this path from BOTH setup and hold analysis, so there is nothing to calculate.
>
> Run: report for config path -> "No paths found."

Data path U_DATA_L -> U_DATA_C - Setup check (report_timing -late) - main check of this experiment

Formula: Setup slack = (capture edge + L_cap - t_setup) - (launch edge + L_lau + t_clk->q + t_data)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	6 x BUF_X1 = 6 x 0.50	3.00
= Data arrival time	**sum of the lines above**	**3.20**
Data required time		
Capture clock edge	next rising edge = T	5.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
- Library setup time	Liberty DFFPOS_X1: setup_rising	-0.20
= Data required time	**sum of the lines above**	**4.80**
Setup slack	**Required - Arrival = 4.80 - 3.20**	**+1.60 MET**

Data path U_DATA_L -> U_DATA_C - Hold check (report_timing -early)

Formula: Hold slack = (launch edge + L_lau + t_clk->q + t_data) - (hold capture edge + L_cap + t_hold)

Quantity	Where the value comes from	Value (ns)
Data arrival time		
Launch clock edge	rising edge at 0	0.00
+ Launch clock network delay	ideal clock (no propagation)	0.00
+ CLK->Q of launch flop	Liberty DFFPOS_X1: CLK->Q (rising_edge)	0.20
+ Data path delay	6 x BUF_X1 = 6 x 0.50	3.00
= Data arrival time	**sum of the lines above**	**3.20**

Quantity	Where the value comes from	Value (ns)
Data required time		
Capture clock edge	same edge	0.00
+ Capture clock network delay	ideal clock (no propagation)	0.00
+ Library hold time	Liberty DFFPOS_X1: hold_rising	0.10
= Data required time	**sum of the lines above**	**0.10**
Hold slack	**Arrival - Required = 3.20 - 0.10**	**+3.10 MET**

Hand calculation vs. run

Path / check	Hand calculation	Run (arrival / required / slack)	Status	Match
Config path U_CFG_L -> U_CFG_C - setup / hold	not timed (false path)	No paths found	NOT TIMED	Yes
data - setup	3.20 / 4.80 / +1.60	3.20 / 4.80 / +1.60	MET	Yes
data - hold	3.20 / 0.10 / +3.10	3.20 / 0.10 / +3.10	MET	Yes

Why this result

- The config path is no longer timed: the report prints a 'no constrained paths' message (OpenSTA: 'No paths found.'). That message is the proof that the exception works.
- The data path is unchanged (+1.60 setup, +3.10 hold) and is the only path in the top-5 report.

Run report - data path, setup (max path, reference run)

```
Startpoint: U_DATA_L (rising edge-triggered flip-flop clocked by CLK)
Endpoint: U_DATA_C (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock CLK (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ U_DATA_L/CLK (DFFPOS_X1)
   0.20    0.20 ^ U_DATA_L/Q (DFFPOS_X1)
   0.50    0.70 ^ b0/Y (BUF_X1)
   0.50    1.20 ^ b1/Y (BUF_X1)
   0.50    1.70 ^ b2/Y (BUF_X1)
   0.50    2.20 ^ b3/Y (BUF_X1)
   0.50    2.70 ^ b4/Y (BUF_X1)
   0.50    3.20 ^ b5/Y (BUF_X1)
   0.00    3.20 ^ U_DATA_C/D (DFFPOS_X1)
           3.20   data arrival time

   5.00    5.00   clock CLK (rise edge)
   0.00    5.00   clock network delay (ideal)
   0.00    5.00   clock reconvergence pessimism
           5.00 ^ U_DATA_C/CLK (DFFPOS_X1)
  -0.20    4.80   library setup time
           4.80   data required time
---------------------------------------------------------
           4.80   data required time
          -3.20   data arrival time
---------------------------------------------------------
           1.60   slack (MET)
```

6.6 Key takeaways

- set_false_path removes setup AND hold checks for the path.
- Use valid start points (clock pins, input ports) and end points (data pins, output ports).
- Always re-report the path after adding the exception to prove it was applied.

6.7 Exercises

1. Replace the false path with set_multicycle_path 2 -setup / 1 -hold on the cfg path. What is the setup slack? (Answer: 9.80 - 6.20 = +3.60.)
2. Change -from to [get_pins U_CFG_L/Q] and read the warning in the log.